Dual Meixner–Pollaczek Functions and a Matching Matrix over a Bessel Ring

Robin Langer and Claude Code

August 2026

Abstract

The Meixner–Pollaczek weight $W(x) = |\Gamma(\frac{1}{4} + ix)|^2$ does not admit a nice square root: $\sqrt{W}$ has branch points at $x = \pm i/4$, and the resulting matching matrix (companion paper) has no algebraic structure. But W factors *analytically*, $W = \Gamma(\frac{1}{4} + ix) \cdot \Gamma(\frac{1}{4} - ix)$, giving a *dual pair* of Meixner–Pollaczek functions $F_m = \Gamma(\frac{1}{4} + ix)\widehat{P}_m$ and $G_m = \Gamma(\frac{1}{4} - ix)\widehat{P}_m$, biorthonormal under the bilinear pairing $\int fg\,dx$. Projecting F_m onto the polynomial basis in $L^2(W)$ yields matching coefficients c_{mk} obeying a phase law $c_{mk} = (-i)^{|m-k|}\,|c_{mk}|$, and — the main observation — after rescaling by $2^m m!\,2^k k!$ every entry lies in the rank-two lattice $\mathbb{Z}A \oplus \mathbb{Z}B$ over the two Bessel constants

$$A = \sqrt{2e}\,K_0(\tfrac{1}{2}), \qquad B = \sqrt{2e}\,K_1(\tfrac{1}{2}).$$

The integrality lost with $\sqrt{W}$ is recovered by the dual pair, with two transcendentals replacing the Hermite matrix's single constant $\sqrt{2\pi/3}$. The integer pairs grow while the entries shrink: each row furnishes rational approximations to $K_1(\frac{1}{2})/K_0(\frac{1}{2})$.

1 The Dual Pair

Let $P_n = P_n^{(1/4)}(\,\cdot\,;\frac{\pi}{2})$ be the Meixner–Pollaczek polynomials, orthogonal on $\mathbb{R}$ with respect to $W(x) = |\Gamma(\frac{1}{4} + ix)|^2$, and $\widehat{P}_n = P_n/\sqrt{h_n}$ the orthonormal family, $h_0 = \int W = \sqrt{2}\,\pi^{3/2}$.

The Gaussian weight factors as $e^{-t^2} = e^{-t^2/2} \cdot e^{-t^2/2}$: two equal positive halves, which is why a single orthonormal basis (the Hermite functions) suffices there. The Meixner–Pollaczek weight refuses this splitting — $\sqrt{W}$ has branch points at $\pm i/4$ — but factors into an analytic function and its reflection:

$$W(x) = \Gamma(\tfrac{1}{4} + ix) \cdot \Gamma(\tfrac{1}{4} - ix).$$

Definition 1.1. The *dual Meixner–Pollaczek functions* are

$$F_m(x) = \Gamma(\tfrac{1}{4} + ix)\,\widehat{P}_m(x), \qquad G_m(x) = \Gamma(\tfrac{1}{4} - ix)\,\widehat{P}_m(x).$$

F_m is meromorphic with poles only at $x = i(\frac{1}{4} + n)$, $n \in \mathbb{N}$ (upper half-plane); G_m has the reflected poles. Under the *bilinear* pairing $\langle f, g\rangle = \int_{\mathbb{R}} fg\,dx$ (no conjugate),

$$\langle F_m, G_k\rangle = \int_{\mathbb{R}} \widehat{P}_m \widehat{P}_k\, W\,dx = \delta_{mk}\,:$$

a biorthonormal system, with each family carrying one *whole* Γ-factor — never a half.

Remark 1.2. $F_m = e^{i\theta(x)}\,\Phi_m$ where $\Phi_m = \widehat{P}_m\sqrt{W}$ is the "obvious" Meixner–Pollaczek function and $e^{i\theta(x)} = \Gamma(\frac{1}{4}+ix)/|\Gamma(\frac{1}{4}+ix)|$ is a unimodular phase (the Γ-part of the Riemann–Siegel theta function). Multiplication by $e^{i\theta}$ is unitary on $L^2(\mathbb{R})$, so $\{F_m\}$ is also orthonormal in the usual sesquilinear sense, and $G_m = \overline{F_m}$. By Bump–Ng, F_m is (up to constants) the Mellin transform of the even Hermite function ψ_{2m} on the critical line.

2 The Matching Coefficients

Projecting F_m onto the polynomial basis in $L^2(W)$:

$$c_{mk} = \langle F_m, \widehat{P}_k \rangle_W = \int_{\mathbb{R}} \widehat{P}_m \widehat{P}_k \, \Gamma(\tfrac{1}{4} + ix)^2 \, \Gamma(\tfrac{1}{4} - ix) \, dx. \tag{1}$$

The coefficients of G_m are the complex conjugates. Three whole Γ-factors appear — no branch points anywhere.

Proposition 2.1 (Checkerboard)**.** *$c_{mk} = c_{km}$, and $x \mapsto -x$ gives $c_{mk} = (-1)^{m+k}\,\overline{c_{mk}}$: entries with $m + k$ even are real, entries with $m + k$ odd are purely imaginary. There is no parity vanishing — the matrix is full.*

Observation 2.2 (Phase law)**.** *For all computed entries ($m, k \le 11$),*

$$c_{mk} = (-i)^{\,|m-k|} \, |c_{mk}|.$$

The phase depends only on $|m - k|$ — an echo of the Fourier eigenvalues $(-i)^m$ of the Hermite functions on the Mellin side.

Table 1 gives the upper-left 10×10 corner (`mp_dual_matching.py`, 40-digit quadrature, cache `mp_dual_matching.pkl`).

Table 1: c_{mk} for $0 \le m, k \le 9$. Real for $m + k$ even, imaginary for $m + k$ odd; the phase is $(-i)^{|m-k|}$.

$m\backslash k$	0	1	2	3	4	5	6	7	8	9
0	2.155	$-.890i$	$-.659$	$.504i$	.425	$-.365i$	$-.324$	$.291i$	.265	$-.244i$
1	$-.890i$	.542	$-.226i$	$-.301$	$.222i$	.223	$-.190i$	$-.182$	$.165i$	.156
2	$-.659$	$-.226i$	.335	$-.084i$	$-.213$	$.125i$	.162	$-.124i$	$-.135$	$.116i$
3	$.504i$	$-.301$	$-.084i$	.235	$-.039i$	$-.169$	$.076i$	.133	$-.086i$	$-.111$
4	.425	$.222i$	$-.213$	$-.039i$	.175	$-.024i$	$-.139$	$.051i$	.114	$-.062i$
5	$-.365i$	.223	$.125i$	$-.169$	$-.024i$	.140	$-.017i$	$-.117$	$.037i$	.099
6	$-.324$	$-.190i$	.162	$.076i$	$-.139$	$-.017i$	.118	$-.012i$	$-.101$	$.028i$
7	$.291i$	$-.182$	$-.124i$	.133	$.051i$	$-.117$	$-.012i$	.102	$-.009i$	$-.089$
8	.265	$.165i$	$-.135$	$-.086i$	.114	$.037i$	$-.101$	$-.009i$	.089	$-.007i$
9	$-.244i$	.156	$.116i$	$-.111$	$-.062i$	.099	$.028i$	$-.089$	$-.007i$	.080

3 Closed Forms: the Bessel Ring

The base integral has a closed form. The three-Γ Mellin–Barnes integral evaluates in terms of Tricomi's confluent hypergeometric function,

$$\int_{\mathbb{R}} \Gamma(a + ix)\Gamma(b + ix)\Gamma(c - ix) \, dx = 2\pi \, \Gamma(a + c)\Gamma(b + c) \, U(a + c,\, a - b + 1,\, 1),$$

and $U(\frac{1}{2}, 1, z) = e^{z/2} K_0(z/2)/\sqrt{\pi}$ gives, at $a = b = c = \frac{1}{4}$,

$$\int_{\mathbb{R}} \Gamma(\tfrac{1}{4} + ix)^2 \, \Gamma(\tfrac{1}{4} - ix) \, dx = 2\pi^{3/2}\sqrt{e} \; K_0(\tfrac{1}{2}) = 16.97348192607767\ldots$$

(verified to 20 digits). Dividing by $h_0 = \sqrt{2}\,\pi^{3/2}$:

$$c_{00} = \sqrt{2e} \; K_0(\tfrac{1}{2}).$$

Since $(a + ix)\Gamma(a + ix) = \Gamma(a + 1 + ix)$, every polynomial moment of the kernel reduces to three-Γ integrals with integer-shifted parameters, hence to *contiguous* U-values — all expressible over the two generators $U(\frac{1}{2}, 1, 1)$ and $U(\frac{3}{2}, 1, 1)$, i.e. over

$$A = \sqrt{2e}\, K_0(\tfrac{1}{2}) = 2.155416\ldots, \qquad B = \sqrt{2e}\, K_1(\tfrac{1}{2}) = 3.862231\ldots$$

Observation 3.1 (Integrality)**.** *Define the rescaled raw entries*

$$T_{mk} = \frac{2^m m!\, 2^k k!}{h_0} \int_{\mathbb{R}} P_m P_k\, \Gamma(\tfrac{1}{4} + ix)^2\, \Gamma(\tfrac{1}{4} - ix)\, dx.$$

Then (dropping the factor i when $m + k$ is odd) every computed entry satisfies

$$T_{mk} \in \mathbb{Z}A \oplus \mathbb{Z}B,$$

confirmed by PSLQ at 40 digits for all $m \le 3$, $k \le 5$ except $(3, 5)$, where the integer coefficients exceed the search bound.

Table 2: Exact entries $T_{mk} = \mathfrak{a}_{mk}A + \mathfrak{b}_{mk}B$ recovered by PSLQ.

(m,k)	T_{mk}	(m,k)	T_{mk}
$(0,0)$	A	$(1,3)$	$2575A - 1440B$
$(0,1)$	$3A - 2B$	$(1,4)$	$39411A - 21978B$
$(0,2)$	$12B - 23A$	$(1,5)$	$394932B - 707389A$
$(0,3)$	$138B - 241A$	$(2,2)$	$2369A - 1320B$
$(0,4)$	$3265A - 1800B$	$(2,3)$	$33743A - 18834B$
$(0,5)$	$52907A - 29706B$	$(2,4)$	$317892B - 569719A$
$(1,1)$	$12B - 21A$	$(2,5)$	$6190602B - 11092229A$
$(1,2)$	$114B - 205A$	$(3,3)$	$295572B - 529549A$
		$(3,4)$	$5399634B - 9675553A$

Remark 3.2 (Diophantine near-cancellation)**.** The integer pairs grow rapidly while $T_{mk} \to 0$, so each entry is a near-cancellation: $\mathfrak{b}_{mk}/\mathfrak{a}_{mk} \approx -A/B$, i.e. the rows generate rational approximations to

$$\frac{B}{A} = \frac{K_1(\frac{1}{2})}{K_0(\frac{1}{2})} = 1.791825\ldots \qquad \Big(\text{e.g. } \tfrac{52907}{29706} = 1.78103\ldots\Big).$$

This is the Padé/continued-fraction mechanism behind irrationality results for Bessel ratios (the Gauss continued fraction), here arising spontaneously from an approximation problem in $L^2(W)$.

4 Comparison with the Self-Dual Matrix

Against the $\sqrt{W}$ matching matrix of the companion paper:

	self-dual ($\sqrt{W}$)	dual pair (Γ^+, Γ^-)
Γ-count in integrals	3/2 (branch points)	3 (poles only)
parity	$b_{mk} = 0$, $m + k$ odd	full matrix, real/imaginary
algebraic structure	none (PSLQ negative)	$\mathbb{Z}A \oplus \mathbb{Z}B$
base constant	$20.9154\ldots$ (no closed form)	$2\pi^{3/2}\sqrt{e}\, K_0(\frac{1}{2})$
biorthogonal expansion	—	$\langle F_m, G_k\rangle = \delta_{mk}$ (exact)

The decay of $|c_{0k}|$ remains slow (the pole of F_m at $x = i/4$ is still at finite distance; local log-log slope ≈ 0.7 at $k = 11$), so the dual pair does not speed up L^2 truncation. What it changes is the *arithmetic*: the coefficients are now exact elements of a rank-two Bessel lattice, computable by residue calculus rather than quadrature, and the biorthogonal expansion itself (F against G) is exact at every order.

In the Hermite case a single transcendental $\sqrt{2\pi/3}$ carried an integer matrix with a Wick matching interpretation, and the modified Bessel function I_0 appeared in the diagonal's generating function. Here *two* transcendentals $K_0(\frac{1}{2})$ and $K_1(\frac{1}{2})$ carry a pair of integer matrices

$(\mathfrak{a}_{mk}, \mathfrak{b}_{mk})$. Open: a combinatorial law for $(\mathfrak{a}, \mathfrak{b})$ (the denominators $2^m m!\, 2^k k!$ and the diagonal pattern suggest a weighted-path model on the Jacobi matrix), a proof of the phase law (Observation 2.2), and the exact decay exponent.